

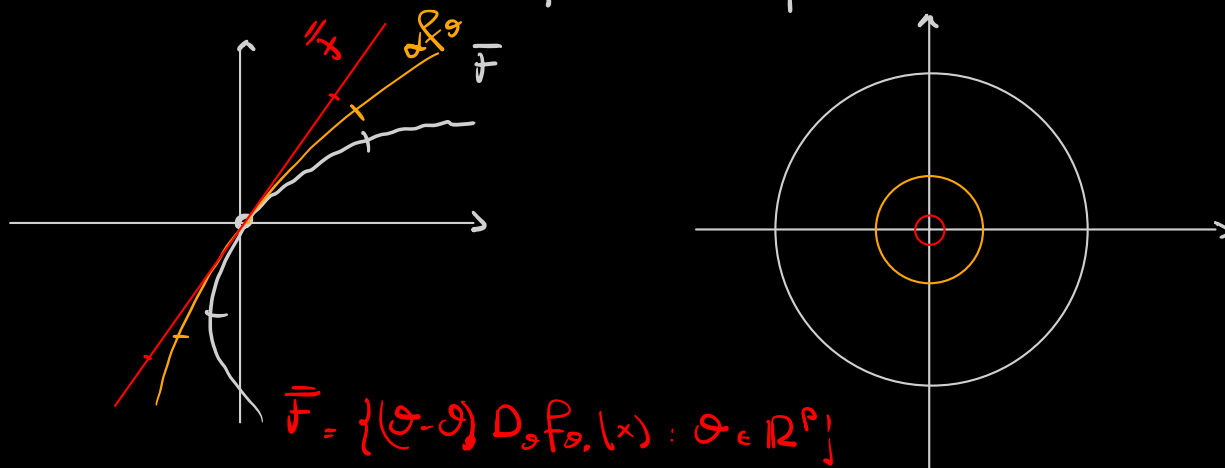
Question: How do NN (NTK) behave as  $n \rightarrow \infty$  during training?

Heuristic discussion: consider rescaling a model  $f_\theta$  as

$$f_\theta \rightarrow \alpha f_\theta \quad \text{for } \alpha \gg 1.$$

To make sense of initialization, assume  $f_{\theta_0} \equiv 0$ .

Then, the model in function space looks like



$$\bar{F} = \{(\theta - \theta_0) D_\theta f_{\theta_0}(x) : \theta \in \mathbb{R}^p\}$$

$$\Rightarrow \bar{F}_\alpha \xrightarrow[\alpha \rightarrow \infty]{} \bar{F}$$

It turns out that neural networks in (NTK) are in this regime

None of this is rigorous, but the intuition can help us construct a proof

Consider the MSE  $\hat{R}(\theta) = \frac{1}{2n} \sum (y_i - f_\theta(x_i))^2 = \frac{1}{2n} \|\vec{y} - \vec{F}(\theta)\|_2^2$

where  $\vec{F}_n(\theta) = (f_\theta(x_1) \dots f_\theta(x_n))$ .

The GF evolution of the model is given by

$$\frac{d}{dt} \theta_t = \frac{1}{n} D_{\theta} \vec{F}_n(\theta) (\vec{y} - \vec{F}_n(\theta)). \quad D_{\theta} \vec{F}_n \in \mathbb{R}^{n \times p}$$

we want to compare this with the evolution of the linearised model in  $\bar{F} := \{ \bar{F} = f_{\theta_0}(x) + D_{\theta} f_{\theta_0}(x) (\theta - \theta_0) : \theta \in \mathbb{R}^p \}$ .

$$\frac{d}{dt} \bar{\theta}_t = \frac{1}{n} D_{\theta} \bar{F}_n(\theta_0) \cdot (\bar{y} - \bar{F}_n(\theta_0) - D_{\theta} \bar{F}_n(\theta_0) (\bar{\theta}_t - \theta_0))$$

which is the GF for the ish

$$\hat{R}_{\text{lin}}(\bar{\theta}) = \frac{1}{2n} \| \bar{y} - \bar{F}_n(\theta_0) - D_{\theta} \bar{F}_n(\theta_0) (\bar{\theta} - \theta_0) \|_2^2$$

To quantify deviations from linearity we consider

$$\text{Lip}(D_{\theta} \bar{F}_n) = \sup_{\theta_1 \neq \theta_2} \frac{\| D_{\theta} \bar{F}_n(\theta_1) - D_{\theta} \bar{F}_n(\theta_2) \|}{\| \theta_1 - \theta_2 \|}$$

where  $\|A\| = \sup_{v \in \mathbb{R}^p, \|v\|_2=1} \|Av\|_2$  and the population version

$$\text{Lip}(D_{\theta} f(\theta)) = \frac{\| D_{\theta} f(\theta_1) - D_{\theta} f(\theta_2) \|}{\| \theta_1 - \theta_2 \|_2}$$

where  $\|D_{\theta} f(\theta)\| = \sup_{v \in \mathbb{R}^p} \frac{\|Df(\theta)v\|_{L^2(P)}}{\|v\|_2}$  for  $Df: \mathbb{R}_+^p \rightarrow \mathcal{L}(P)$

Thm 1 Let  $\sigma_{\max} = \sigma_{\max}(Df_u(\vartheta_0))$ ,  $\sigma_{\min} = \sigma_{\min}(Df_u(\vartheta_0))$

Assume  $\text{Lip } D_{\vartheta} \bar{f}_u \|y - \bar{f}_u(\vartheta_0)\| \leq \frac{1}{4} \sigma_{\min}^2$

Then for all  $t \geq 0$ :

$$1) \quad \hat{R}(\vartheta_t) \leq \hat{R}(\vartheta_0) e^{-\lambda_0 t} \quad \lambda_0 = \sigma_{\min}^2 / 2n$$

$$2) \|f(\vartheta_t) - \bar{f}(\vartheta_t)\|_{L^2(\rho)} \leq \left( \frac{\text{Lip}(Df)}{\sigma_{\min}^2} + \frac{\|Df(\vartheta_0)\| \text{Lip } Df_u}{\sigma_{\min}^5 / \sigma_{\max}^2} \right) \|y - \bar{f}_u(\vartheta_0)\|^2$$

Assumption: Training only happens in the hidden layer,  
furthermore  $\vartheta_j \stackrel{\text{iid}}{\sim} \text{Unif}\{-1, 1\}$ .

Lemma 2.  $\text{Lip}(Df) = O(\sqrt{\frac{d}{m}})$   $\text{Lip}(Df_u) = O(\sqrt{\frac{d}{m}}(\sqrt{n} + \sqrt{d}))$

$$\bullet \sigma_{\min} = O(\sqrt{d}) \quad \sigma_{\max} = O(\sqrt{n} + \sqrt{d})$$

$$\bullet \|y - f_u(\vartheta_0)\| = O(\sqrt{n}) \quad \|Df(\vartheta_0)\| = O(1)$$

Thm 3 Let  $(x_i, y_i)_{i=1}^n$ :  $x_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \mathbb{I}_d)$ ,  $y_i$   $\mathbb{R}^2$ -sub-gaussian.

Let  $\sigma = \sum_{e \in \mathcal{E}_0} \mu_e(\sigma) h_e$  for  $h_e$  the Hermite polynomials, and  $\mu_e(\sigma) \neq 0 \quad \forall \quad e \in \mathcal{E}_0$ . Further let  $u \geq c \log n$ ,  $n \leq d^{e_0}$ .

Then w.p. at least  $1 - \exp(-u/c)$ , for all  $t \geq 0$  we have

$$1) \quad \hat{R}(\vartheta_t) \leq \hat{R}(\vartheta_0) e^{-\lambda_* t} \quad \lambda_* = C_1 \frac{d}{u}$$

$$2) \quad \|f_{\vartheta_t} - \bar{f}_{\vartheta_t}\|_{L^2(\rho)} \leq C_1 \left( \sqrt{\frac{u^2}{nd}} + \sqrt{\frac{u^5}{nd^4}} \right) = O\left(\frac{1}{\sqrt{m}}\right)$$

Constants  
are uniform  
in  $n$

Proof of Thm 1: def.  $y_t = f_u(\theta_t)$ ,  $K_t = Df_u(\theta_t) \cdot Df_u(\theta_t)^T \in \mathbb{R}^{n \times n}$

$$\dot{y}_t = Df_u(\theta_t) \dot{\theta}_t = -\frac{1}{u} Df_u(\theta_t) Df_u^T(y_t - y)$$

$$\Rightarrow \frac{d}{dt} \mathcal{R}(\theta_t) = \frac{d}{dt} \frac{1}{2} \|y_t - y\|_2^2 = -\frac{1}{u} (y_t - y)^T K_t (y_t - y)$$

$$\text{Let } r_* = \frac{\sigma_{\min}}{2 \text{Lip}(D_{\theta} f_u)}, \quad t_* = \inf \{t > 0 : \|\theta_t - \theta_0\| \geq r_*\} \text{ such}$$

$$\text{that } \sigma_{\min}(D_{\theta} f_u(\theta_t)) \geq \sigma_{\min}(D_{\theta} f_u(\theta_0)) + D_{\theta} f_u(\theta_t) - D_{\theta} f_u(\theta_0)$$

$$= \sigma_{\min}(D_{\theta} f_u(\theta_0)) - \frac{\|D_{\theta} f_u(\theta_t) - D_{\theta} f_u(\theta_0)\| \|\theta_t - \theta_0\|}{\|\theta_t - \theta_0\|}$$

$$\leq \sigma_{\min}(D_{\theta} f_u(\theta_0)) - \text{Lip}(D_{\theta} f_u) \cdot r_* \leq \frac{\sigma_{\min}}{2}$$

$$\Rightarrow \lambda_{\min}(K_t) = \sigma_{\min}(D_{\theta} f_u)^2 \leq \left(\frac{\sigma_{\min}}{2}\right)^2 \quad \forall t < t_*$$

$$\Rightarrow \forall t < t_* \quad \frac{d}{dt} \frac{1}{2} \|y_t - y\|^2 \leq -\frac{\sigma_{\min}^2}{2u} \cdot \frac{1}{2} \|y_t - y\|^2 \Rightarrow \frac{1}{2} \|y_t - y\|^2 \leq e^{-\frac{\sigma_{\min}^2}{2u} t} \frac{1}{2} \|y_0 - y\|^2$$

From this we can bound  $\|\theta_t - \theta_0\|$ :

$$\|\dot{\theta}_t\|_2 = \frac{1}{u} \|Df_u(\theta_t)^T (y_t - y_0)\|$$

$$\frac{d}{dt} \|y_t - y\| = -\frac{1}{u} \frac{1}{2} \frac{\frac{d}{dt} \|y_t - y\|^2}{\|y_t - y\|} = -\frac{1}{u} \frac{\|D_{\theta} f_u^T (y_t - y)\|^2}{\|y_t - y\|} \leq -\frac{\sigma_{\min}}{2u} \|Df_u^T (y_t - y)\|$$

$$\frac{d}{dt} \|\cdot\| \leq \left| \frac{d}{dt} \|\cdot\| \right|$$

$$\Rightarrow \frac{d}{dt} \left( \|y_t - y\| + \frac{\sigma_{\min}}{2} \|\theta_t - \theta_0\| \right) \leq \frac{d}{dt} \|y_t - y\| + \frac{\sigma_{\min}}{2} \|\dot{\theta}_t\| \leq 0 \quad \forall t < t_*$$

$$\Rightarrow \|\theta_t - \theta_0\| \leq \frac{2}{\sigma_{\min}} \|y - y_t\| \leq \frac{2}{\sigma_{\min}} \|y - y_0\|$$

Now assume that  $t_* < \infty$ , then by assumption

$$\text{Lip}(D_\theta f_u) \|y - y_0\| < \frac{1}{4} \sigma_{\min}^2(D_\theta f_u(\theta_0))$$

so that  $\|\theta_{t_*} - \theta_0\| \leq \frac{2}{\sigma_{\min}} \|y - y_0\| < \frac{1}{2} \sigma_{\min} \cdot \frac{1}{\text{Lip } D_\theta f_u} = r_*$   $\checkmark$

$\Rightarrow t_* = \infty$ .

$$\|f(\theta_*) - \bar{f}(\bar{\theta}_*)\| \leq \underbrace{\|f(\theta_*) - \bar{f}(\theta_*)\|}_{E_1} + \underbrace{\|\bar{f}(\theta_*) - \bar{f}(\bar{\theta}_*)\|}_{E_2}$$

$$\begin{aligned} E_1 &= \left\| \int_0^t \frac{d}{ds} (f(\theta_s) - \bar{f}(\theta_s)) ds \right\| = \left\| \int_0^t (D_\theta f(\theta_s) - D_\theta f(\theta_0)) \dot{\theta}_s ds \right\| \\ &\leq \text{Lip } D_\theta f \underbrace{\sup \|\theta_s - \theta_0\|}_{\leq \frac{2}{\sigma_{\min}} \|y - y_0\|} \underbrace{\int_0^t \|\dot{\theta}_s\| ds}_{\leq \frac{2}{\sigma} \frac{d}{ds} \|y_s - y\|} \leq \text{Lip } D_\theta f \frac{4}{\sigma_{\min}^2} \|y - y_0\|^2 \end{aligned}$$

$$E_2 = \|D_\theta f(\theta_0)(\theta_* - \bar{\theta}_*)\| \leq \|D_\theta f(\theta_0)\| \|\theta_* - \bar{\theta}_*\| \leq 180 \frac{L_u \sigma_{\max}^2}{\sigma_{\min}^5} \|D_\theta f(\theta_0)\| \|y - y_0\|$$

Claim:  $\|\theta_* - \bar{\theta}_*\| \leq 180 \frac{L_u \sigma_{\max}^2}{\sigma_{\min}^5} \|y - y_0\|^2$

Sketch of proof let  $v_+ = y_+ - \bar{y}_+$  so that

$$\begin{aligned} \dot{v}_+ &= \dot{y}_+ - \dot{\bar{y}}_+ = -\frac{1}{u} D_\theta f_u(\theta_+) (D_\theta f_u(\theta_+) (f_u(\theta_+) - y)) \\ &\quad + \frac{1}{u} D_\theta f_u(\theta_0) (D_\theta f_u(\theta_0) (\bar{f}_u(\bar{\theta}_+) - y)) \\ &= -\frac{1}{u} \kappa_+ v_+ - \underbrace{\left( \frac{\kappa_+ - \kappa_0}{u} \right) (\bar{f}_u(\bar{\theta}_+) - y)}_{O(\sigma_{\max}/\sigma_{\min}^3 L_u)} \Rightarrow \text{Gronwall} \quad \square \end{aligned}$$

Proof of Lemma 2:

$$D_{\theta} f_u(\theta)_i = \text{diag}\left(\frac{a}{\sqrt{m}}\right) \cdot \sigma'(w \cdot x_i) x_i$$

$$\Rightarrow \|D_{\theta} f_u(\theta) - D_{\theta} f_u(\theta')\|^2 = \sum_{i=1}^m \left\| \text{diag}\left(\frac{a}{\sqrt{m}}\right) (\sigma'(w \cdot x_i) - \sigma'(w' \cdot x_i)) x_i \right\|^2$$

$$\leq \frac{1}{m} \|a\|_{\infty}^2 \|X\|_2^2 \max_{1 \leq j \leq n} \|\sigma'(w \cdot x_j) - \sigma'(w' \cdot x_j)\|^2$$

$$\leq \frac{1}{m} \underbrace{\|X\|_2^2}_{\sim d} C \|w - w'\|^2$$

$$\Rightarrow \|D_{\theta} f_u(\theta) - D_{\theta} f_u(\theta')\| \leq \sqrt{\frac{d}{m}} \|\theta - \theta'\| C$$

Summary: the model

$$f_{\theta}(x) = \frac{1}{\sqrt{m}} \sum a_f \sigma(w_f \cdot x + b_f)$$

behaves, to first order approximation, as

$$\bar{f}_{\bar{\theta}}(x) = f_{\theta_0}(x) + D_{\theta} f_{\theta_0}(x) (\bar{\theta} - \theta_0)$$

Note that in this case  $D_{\theta} f_{\theta_0}(x)$  can be computed explicitly

$$D_{\theta} f_u(\theta_0) = (D_a f_u(\theta_0), D_w f_u(\theta_0), D_b f_u(\theta_0))$$

$$= \left( \frac{1}{\sqrt{m}} \delta_{ae} \sum_{j=1}^m a_j \sigma(w_j x + b_j), \frac{1}{\sqrt{m}} D_{w_e} \sum a_j \sigma(w_j x + b_j), \frac{1}{\sqrt{m}} \delta_{be} \sum \dots \right)$$

$$= \frac{1}{\sqrt{m}} \left( \sigma(w_e \cdot x_e + b_e) \right)_{e,e}, \left( a_e \sigma'(w_e \cdot x_e + b_e) x_{e,e} \right)_{e,e}$$

$$= \frac{1}{\sqrt{m}} \left( \varphi_{RF}(x_e), \varphi_{NT}(x_e) \right)_e$$

$\mathbb{R}^m$                        $\mathbb{R}^{md}$

The above discussion can be easily extended to UPs of the form

$$f_\sigma(x) = \frac{1}{\sqrt{m_L}} \sum_{l=1}^{m_L} w_{lL}^{(L)} \cdot \sigma \left( \frac{1}{\sqrt{m_{L-1}}} \sum \dots \sigma \left( \frac{1}{\sqrt{m_1}} \sum w_{j1}^{(1)} x_j + b_{j1} \right) + \dots + b_{d_{L-1}} \right)$$

when  $m_2, \dots, m_L \rightarrow \infty$